Math Kangaroo Practice

Grade 3-4 · 32 questions · standard · seed 11776

Part A: Each question is worth 3 points

1. The numbers 34 and 142 have the same sum of digits (3 + 4 = 7 and 1 + 4 + 2 = 7). What is the first number greater than 220 whose sum of digits is the same as for 220?

A. 221 ***B.*** 229 ***C.*** 230 ***D.*** 233 ***E.*** 301

2. Mia writes all the numbers from 95 to 107.
How many digits does Mia write?

A. 34 ***B.*** 35 ***C.*** 39 ***D.*** 40 ***E.*** 42

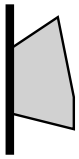
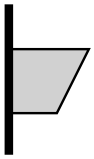
3. Lucas, Ben, Hana and Omar each play a different instrument: the piano, the drum, the trumpet and the harp. Lucas plays an instrument with strings. Hana blows into the instrument. Ben presses keys.
Who plays the harp?

A. Hana ***B.*** Omar ***C.*** Lucas
D. Ben ***E.*** Impossible to determine

4. To write all the room numbers in a hotel, starting from 1, exactly 37 digits are needed.
How many rooms are there?

A. 15 ***B.*** 18 ***C.*** 23 ***D.*** 32 ***E.*** 37

5. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Sara draws the shadow at five different moments.
Which picture **cannot** be right unless the flag is torn?

A.  ***B.*** ***C.*** ***D.***  ***E.***

6. These diagrams follow a pattern: a black square with a white border around it. How many white cells will the next diagram have?

8 white cells

12 white cells

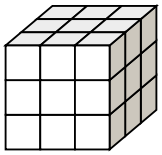
16 white cells

A. 20 ***B.*** 18 ***C.*** 17 ***D.*** 16 ***E.*** 15

7. The digits from 1 to 9 are written on 9 cards. Alex has 7, 6 and 2; Nina has 1, 5 and 8; Martha has 3, 4 and 9. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.
Who **cannot** get 15 as the result?

A. Alex ***B.*** Nina ***C.*** Martha
D. All can get 15 ***E.*** Nina and Martha

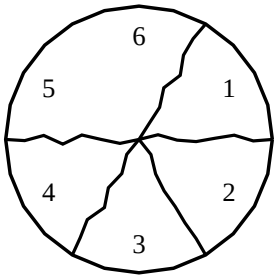
8. A cube with sides of 3 cm is painted grey all over and then cut into small cubes with sides of 1 cm.



How many of the small cubes have exactly 2 painted faces?

A. 4 ***B.*** 6 ***C.*** 8 ***D.*** 12 ***E.*** 36

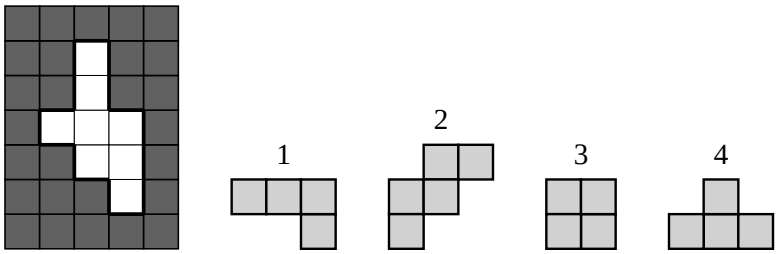
9. A round plate with the numbers 1 to 6 around its edge is cracked into 5 pieces as shown.



Which piece has the smallest sum of numbers?

A. the piece with the 2 ***B.*** the piece with the 6 ***C.*** the piece with the 3
D. the piece with the 1 ***E.*** the piece with the 4

10. Which two of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2 ***B.*** 1+3 ***C.*** 1+4 ***D.*** 2+3 ***E.*** 3+4

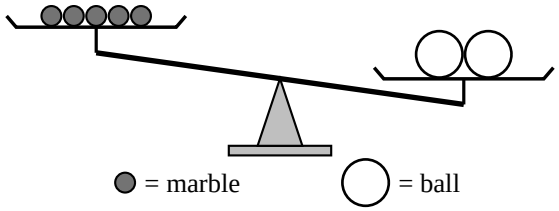
11. How many letters do the words TEACHER and GARDEN have in common?

A. 7 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 3

12. Five ribbons have these lengths.
Which ribbon is the longest?

A. 6600 mm ***B.*** 6 m 20 cm ***C.*** 570 cm ***D.*** 55 dm ***E.*** 61 dm

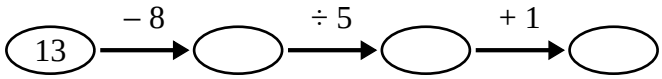
13. On one plate of a balance there are 5 marbles and on the other there are 2 balls. When we put one more ball, exactly like the others, on the plate with the marbles, the balance is level.



One ball weighs the same as how many marbles?

A. 10 marbles ***B.*** 9 marbles ***C.*** 7 marbles ***D.*** 6 marbles ***E.*** 5 marbles

14. Follow the arrows and do each calculation in turn.



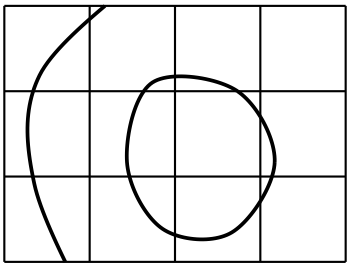
Which number comes out at the end?

A. 104 ***B.*** 13 ***C.*** 3 ***D.*** 2 ***E.*** 1

15. The odometer of a bike shows 585 km. This number reads the same forwards and backwards.
After how many more kilometres will this happen again?

A. 1 ***B.*** 10 ***C.*** 11 ***D.*** 20 ***E.*** 31

16. Some paths go across a park. Here is the whole picture.

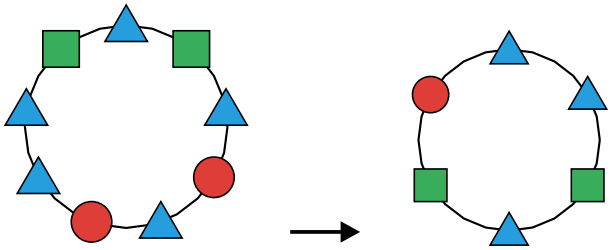


Which of these squares is **not** a piece of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

Part B: Each question is worth 4 points

17. Ben takes away 3 neighbouring stones from the bracelet on the left. Then the bracelet looks like the one on the right. Which stones did Ben take away?

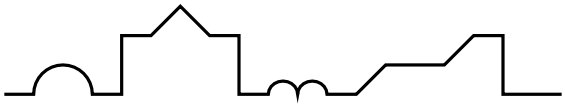


A. ***B.*** ***C.*** ***D.*** ***E.***

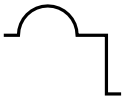
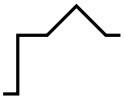
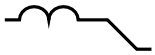
18. Mia builds a brick 5 cubes long and 3 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Mia used exactly 9 blue cubes.
How many cubes deep is the brick?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

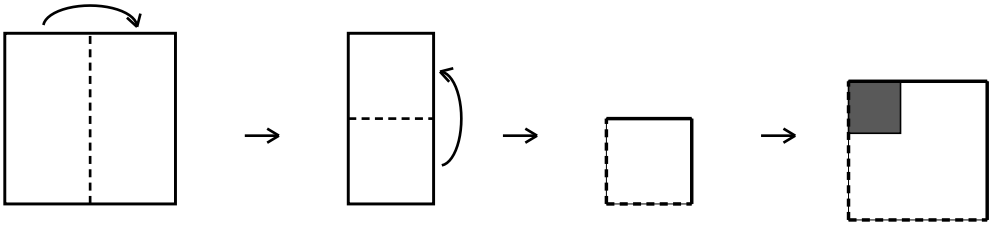
19. Lena cut a strip of paper along this line.



Which piece **can** come from the cut line?

A. ***B.***  ***C.***  ***D.*** ***E.*** 

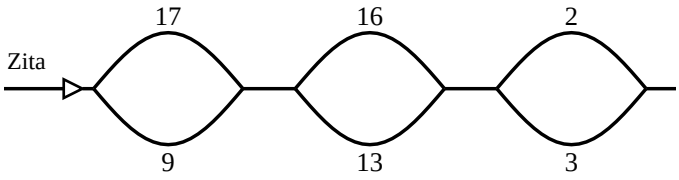
20. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). The dark part is cut off and falls on the table. It is four layers of paper.



How many separate pieces of paper are on the table?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 8

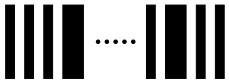
21. Zita walks along the road from left to right and adds up the numbers along the way.



Which total can Zita **not** get?

A. 24 ***B.*** 27 ***C.*** 29 ***D.*** 32 ***E.*** 33

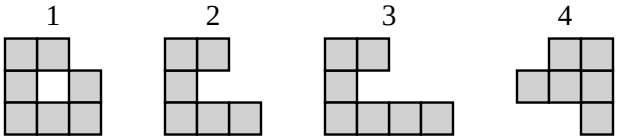
22. A bar code is made of 23 black and white bars taking turns, with a black bar at each end. Each black bar is wide or narrow. There are 10 more white bars than wide black bars.



How many wide black bars are there?

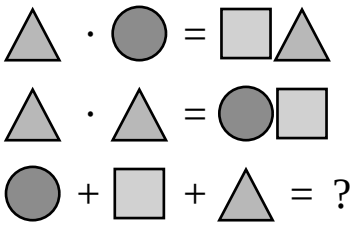
A. 21 ***B.*** 11 ***C.*** 2 ***D.*** 1 ***E.*** 0

23. Which two of these pieces fit together to make a rectangle? Pieces may be turned and flipped over.



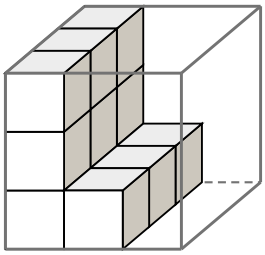
A. 1 and 4 ***B.*** 3 and 4 ***C.*** 2 and 4 ***D.*** 2 and 3 ***E.*** 1 and 2

24. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Tom lays out the cards so that the calculations are correct. What do you get if you add the digits of all three shapes?



A. 14 ***B.*** 15 ***C.*** 16 ***D.*** 17 ***E.*** 18

25. Leo puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Now Leo wants to cover the whole bottom of the box with one layer of cubes.



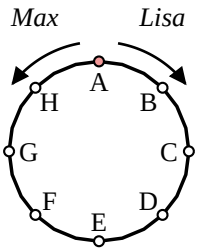
How many more cubes are needed for that?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 15

26. Tom was born on October 25, 2012 and Ben on November 6, 2012.
By how many days is Tom older than Ben?

A. 19 ***B.*** 13 ***C.*** 12 ***D.*** 11 ***E.*** 9

27. Lisa and Max start at the same time from point A of the circle. Lisa runs clockwise, Max runs anticlockwise (see picture). They meet for the first time at point C, then at E, at G and then again at point A. How many times has Max run around the circle by then?



A. 1 time ***B.*** 2 times ***C.*** 3 times ***D.*** 4 times ***E.*** 6 times

28. 51 children stand in a row. Each child has a vest and a backpack. The vests repeat like this: yellow, green, yellow, green, ... The backpacks repeat like this: red, purple, brown, red, purple, brown, ... How many children have a **green** vest **and** a **purple** backpack?

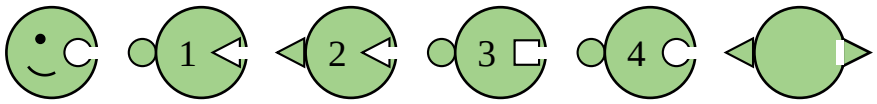
A. 7 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 25

29. Four children, Iris, Clara, Ben and Grace, sit on a bench. Iris always sits at the left end and Clara always sits at the right end.
In how many different ways can the children sit on the bench?

A. 1 ***B.*** 2 ***C.*** 4 ***D.*** 6 ***E.*** 24

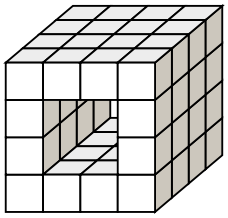
Part C: Each question is worth 5 points

30. Tom has these pieces: a head, a tail and 4 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3 or 4 parts, then the tail. Each piece can be used only once. Tom has already put part 4 right after the head. In how many different ways can Tom finish the caterpillar?



A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

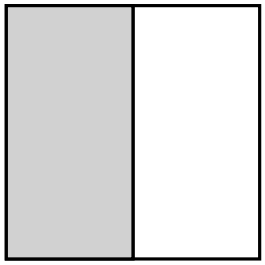
31. Mia's tunnel is 4 cubes long. The hole goes straight through. Mia makes it longer by adding more cubes at the back in the same pattern, until it is 8 cubes long.



How many cubes are in the long tunnel?

A. 128 ***B.*** 108 ***C.*** 100 ***D.*** 96 ***E.*** 48

32. Mia cuts a square chocolate bar into 2 equal strips as shown. The perimeter of each strip is 18 cm.



How long is the side of the square?

A. 2 cm ***B.*** 3 cm ***C.*** 4 cm ***D.*** 5 cm ***E.*** 6 cm

Answer Key

1.	**E**	**2.**	**A**	**3.**	**C**	**4.**	**C**	**5.**	**E**	**6.**	**A**
7.	**B**	**8.**	**D**	**9.**	**D**	**10.**	**C**	**11.**	**E**	**12.**	**A**
13.	**E**	**14.**	**D**	**15.**	**B**	**16.**	**A**	**17.**	**A**	**18.**	**A**
19.	**C**	**20.**	**B**	**21.**	**C**	**22.**	**D**	**23.**	**C**	**24.**	**E**
25.	**D**	**26.**	**C**	**27.**	**C**	**28.**	**C**	**29.**	**B**	**30.**	**A**
31.	**D**	**32.**	**E**								

Solutions

1. E (301) — 2 + 2 + 0 = 4. Going up from 220, the first number with digit sum 4 is 301 (3 + 0 + 1 = 4). [set-03/by_level/Grade_03-04/2007 Q4 · next_same_sum]

2. A (34) — There are 13 numbers: 5 × 2 + 8 × 3 = 34 digits. [set-03/by_level/Grade_03-04/2007 Q23 · across_hundred]

3. C (Lucas) — The only possible matching: Lucas – harp; Ben – piano; Hana – trumpet; Omar – drum. [set-03/by_level/Grade_03-04/2002 Q18 · who_has]

4. C (23) — 1 to 9 use 9 digits. The other 37 – 9 = 28 digits belong to two-digit numbers: 28 ÷ 2 = 14 numbers, from 10 to 23. So there are 23. [set-03/by_level/Grade_03-04/2003 Q16 · pages_from_digits]

5. E — The side sewn to the pole always shows at its full length. In picture E part of the pole is bare, so a piece of the flag is missing. [set-03/by_level/Grade_03-04/2002 Q15 · torn_flag]

6. A (20) — The border of an 6 × 6 square has 6 + 6 + 4 + 4 = 20 cells (or 6 × 6 – 4 × 4 = 20). [set-03/by_level/Grade_03-04/2007 Q21 · border_next]

7. B (Nina) — Alex: 2 + 6 + 7 = 15; Martha: 3 × (9 – 4) = 15. Nina: with 1, 5 and 8 nothing gives 15 (the closest are 14 = 1 + 5 + 8 and 32 = (5 – 1) × 8). [set-03/by_level/Grade_03-04/2002 Q21 · who_cannot]

8. D (12) — They sit on the edges, but not at the corners. 12 edges with 3 – 2 = 1 such cubes each: 12 × 1 = 12. [set-03/by_level/Grade_03-04/2007 Q17 · two_faces]

9. D (the piece with the 1) — The sums are: 1 = 1, 2 = 2, 3 = 3, 4 = 4, 5+6 = 11. [set-03/by_level/Grade_03-04/2002 Q11 · biggest_piece]

10. C (1+4) — The empty area has 8 squares. Pieces have 1: 4, 2: 5, 3: 4, 4: 4 squares. Only pieces 1 and 4 also fit together in the right shape. [set-03/by_level/Grade_03-04/2003 Q18 · two_pieces]

11. E (3) — Check each different letter of TEACHER: only A, E and R are also in GARDEN, so 3. A letter that appears twice still counts once. [set-03/by_level/Grade_03-04/2007 Q3 · common_letters]

12. A (6600 mm) — In cm: 6600 mm = 660 cm, 6 m 20 cm = 620 cm, 570 cm = 570 cm, 55 dm = 550 cm, 61 dm = 610 cm. The longest is 6600 mm. [set-03/by_level/Grade_03-04/2003 Q10 · longest]

13. E (5 marbles) — Take 1 ball off both plates: 5 marbles balance 2 – 1 = 1 ball. So one ball = 5 marbles. [set-03/by_level/Grade_03-04/2002 Q9 · one_more_balances]

14. D (2) — 13 – 8 = 5, 5 ÷ 5 = 1, 1 + 1 = 2. [set-03/by_level/Grade_03-04/2007 Q5 · forwards]

15. B (10) — The next such number is 595, so 595 – 585 = 10. [set-03/by_level/Grade_03-04/2002 Q24 · next_palindrome]

16. A — Find each square in the picture by the places where its lines reach the edges. Squares B, C, D, E can be found; square A is a piece mirrored and turned, so it is not in the picture like that. [set-03/by_level/Grade_03-04/2002 Q1 · not_a_piece]

17. A — Count each shape before and after: circles 2 → 1, squares 2 → 2, triangles 5 → 3. So Ben took away 1 circle and 2 triangles. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · which_removed]

18. A (5) — The blue inside is 3 long and 1 tall, so one layer of it has 3 × 1 = 3 cubes. 9 ÷ 3 = 3 layers of blue, plus the red front and back: 5. [set-03/by_level/Grade_03-04/2003 Q9 · how_deep]

19. C — Only this piece is found in the line. The others don't fit — A: it is a mirror image of a part of the line; B: it steps down where the line steps up; D: it steps up where the line steps down; E: its slope goes the wrong way. [set-03/by_level/Grade_03-04/2002 Q7 · can_belong]

20. B (2) — The cut is at the middle of a paper side: the layers are joined in pairs along one fold, so the cut gives 2 pieces (two notches). [set-03/by_level/Grade_03-04/2007 Q24 · count_pieces]

21. C (29) — 24 = 9 + 13 + 2; 27 = 9 + 16 + 2; 32 = 17 + 13 + 2; 33 = 17 + 13 + 3. No route gives 29. [set-03/by_level/Grade_03-04/2007 Q1 · impossible_total]

22. D (1) — There are 12 black and 11 white bars. Wide: 11 – 10 = 1. [set-03/by_level/Grade_03-04/2003 Q17 · wide_bars]

23. C (2 and 4) — Pieces 2 and 4 have 12 squares together and fit into a 4 × 3 rectangle. No other pair does. [set-03/by_level/Grade_03-04/2007 Q8 · two_make_rectangle]

24. E (18) — Try the possible digits (start with the multiplication: which times-table result has these digits?) and check every calculation: circle = 6, square = 4, triangle = 8. 6 + 4 + 8 = 18. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · times]

25. D (3) — The bottom has 3 × 3 = 9 places. 6 of them already have a column of cubes. 9 – 6 = 3. [set-03/by_level/Grade_03-04/2007 Q14 · bottom_layer]

26. C (12) — October has 31 days, so from October 25 to the end of the month is 6 days, then 6 more days to November 6, 2012: 12 days. [set-03/by_level/Grade_03-04/2007 Q12 · days_between]

27. C (3 times) — Each time, Lisa runs 2 of the 8 parts and Max runs the other 6. They meet 4 times, so Max runs 4 × 6 = 24 parts = 3 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · skip_points]

28. C (9) — The patterns start again together every 6 children; in each block child number 2 matches. 51 = 8 × 6 + 3: 8 full blocks give 8, and in the 3 children left over child 50 matches too. 8 + 1 = 9. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · leftover]

29. B (2) — Only the 2 children in the middle can change places: 2 × 1 = 2 ways. [set-03/by_level/Grade_03-04/2002 Q22 · ends_fixed]

30. A (2) — Look only at the branch that starts with part 4: H–4–1–T, H–4–1–2–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · start_given]

31. D (96) — Each slice of the tunnel has 4 × 4 – 2 × 2 = 12 cubes. 8 slices: 12 × 8 = 96. [set-03/by_level/Grade_03-04/2002 Q20 · longer_tunnel]

32. E (6 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 18 ÷ 6 = 3, so the side is 2 × 3 = 6 cm. [set-03/by_level/Grade_03-04/2002 Q6 · strips_to_square]